

Morning Paper Report - Tuesday, June 16, 2026

Recent, non-repeated papers matched to visual working memory, visual search, modeling, working-memory representation, neural-network memory models, AI for human memory, and egocentric predictive-coding memory themes.

1. Decoding Color Visual Working Memory from EEG Signals Using Graph Convolutional Neural Networks

Xiaowei Che, Yuanjie Zheng, Xin Chen, Suta Song, Shouxin Li
International Journal of Neural Systems, 2021

Source: Semantic Scholar; matched terms: visual working memory, working memory, neural network, neural networks

Link: <https://www.semanticscholar.org/paper/934dbecf73f0476375c3a698e68cc00d881ac249>

Goal: Color has an important role in object recognition and visual working memory (VWM)

Method: Here, we constructed an EEG color graph convolutional network model (ECo-GCN) to decode colors during different VWM stages

Results: Recently, several studies showed that color could be decoded from scalp electroencephalogram (EEG) signals during the encoding stage of VWM, which process visible information with strong neural coding

Conclusion: The results suggest that EEG signals during the maintaining stage may be more sensitive than behavioral measurement to predict the VWM performance of human, and ECo-GCN provides an effective approach to explore human cognitive function.

2. Convolutional neural networks uncover the dynamics of human visual memory representations over time.

Eden Zohar, Stas Kozak, Dekel Abeles, M. Shohar, N. Censor
Cerebral Cortex, 2024

Source: Semantic Scholar; matched terms: visual working memory, working memory, neural network, neural networks

Link: <https://www.semanticscholar.org/paper/d64a6dbb0927b09de0672549dc8a75d50320706a>

Goal: The ability to accurately retrieve visual details of past events is a fundamental cognitive function relevant for daily life

Method: Visual memory strength was measured with distractors selected based on their shared similarity to the target memory along low or high layers of the convolutional neural network hierarchy

Results: The results show that visual working memory relies similarly on low and high-level visual representations

Conclusion: These findings suggest that visual representations transform from a distributed to a stronger high-level conceptual representation, providing novel insights into the dynamics of visual memory over time.

3. Uncertainty Quantification in Working Memory via Moment Neural Networks

He Ma, Wenlian Lu, Jianfeng Feng
arXiv.org, 2024

Source: Semantic Scholar; matched terms: working memory, precision, neural network, neural networks

Link: <https://www.semanticscholar.org/paper/126390cf202f1e75197fa6695d4040bd8b1bee94>

Goal: Humans possess a finely tuned sense of uncertainty that helps anticipate potential errors, vital for adaptive behavior and survival

Method: Trained on a WM task, the model demonstrates coding precision and uncertainty quantification comparable to human performance

Results: Transferring the MNN's weights to an SNN replicates these results

Conclusion: Furthermore, the study provides testable predictions demonstrating how noise and heterogeneity enhance WM performance, highlighting their beneficial role rather than being mere biological byproducts. These findings offer insights into how the brain effectively manages uncertainty with exceptional accuracy.

4. Enabling Lifelong Learning in AI with Biologically-inspired Neural Networks Based on Short-Term, Working, and Long-Term Memory*

Hanav Modasiya

International Conference on Cognitive Machine Intelligence, 2025

Source: Semantic Scholar; matched terms: working memory, neural network, neural networks, artificial intelligence

Link: <https://www.semanticscholar.org/paper/a8cdce9d7cc1b302d1d5644b25be97979ac16044>

Goal: Achieving lifelong learning, the ability of a learning system to continuously acquire and adapt to changing data, in Artificial Intelligence (AI) is crucial to achieving Artificial General Intelligence (AGI), a hypothetical, human-like version of AI that has endless possibilities in various domains

Method: Currently, the vast majority of models are incapable of exhibiting lifelong learning because they demonstrate catastrophic forgetting

Results: To support that, experiments revealed that the complementary learning nature of the architecture creates an unexpected learning phenomenon and pattern of machine "reasoning" that increases accuracy

Conclusion: To support that, experiments revealed that the complementary learning nature of the architecture creates an unexpected learning phenomenon and pattern of machine "reasoning" that increases accuracy

5. Comprehensive exploration of visual working memory mechanisms using large-scale behavioral experiment

Liqiang Huang

Nature Communications, 2025

Source: Semantic Scholar; matched terms: visual working memory, working memory, neural network, neural networks

Link: <https://www.semanticscholar.org/paper/863a6fb514dceaae9b044d57fd547b020dbdd415>

Goal: Two decades of research on visual working memory have produced substantial yet fragmented knowledge

Method: Drawing on a large-scale behavioral experiment involving 40 million responses to 10,000 color patterns, a quasi-comprehensive exploration model of visual working memory, termed QCE-VWM, is developed

Results: Two decades of research on visual working memory have yielded substantial yet fragmented knowledge. Here, using a large-scale experiment with 40 million responses, the author provides evidence these findings can be understood within a cohesive framework.

Conclusion: Two decades of research on visual working memory have yielded substantial yet fragmented knowledge. Here, using a large-scale experiment with 40 million responses, the author provides evidence these findings can be understood within a cohesive framework.